

Spherical Chamfer Generative Networks: One-Step Continuous Text-Conditional Generation by Joint Hypersphere Matching

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<https://github.com/intexcor/SCGN>

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Abstract

We introduce Spherical Chamfer Generative Networks (SCGN), a simple one-step conditional image generation prototype trained without adversarial losses, diffusion sampling, paired reconstruction, or explicit optimal-transport solvers. The method maps random noise and a continuous conditioning vector directly to an image in a single forward pass. Its core constraint is geometric: generated and real images are projected onto a fixed-radius hypersphere, and training minimizes a bidirectional Chamfer objective in a joint space formed by image direction and condition direction. The hyperspherical projection prevents the trivial low-variance mean solution that appears under naive pixel matching, while the joint image-condition nearest-neighbor loss encourages conditional routing without class-classification losses. We report preliminary qualitative results on MNIST, Fashion-MNIST, CIFAR-10, and a synthetic text-encoder mode in which each image receives a unique prompt embedding. This technical report is intended to document the method and its initial empirical behavior; large-scale evaluation and comparisons are left for future work.

1 Introduction

Modern image generators are typically trained with one of several powerful but complex mechanisms: adversarial discrimination [2], multi-step diffusion or score-based sampling [3, 6], flow or transport-based objectives [4, 5], or explicit optimal-transport approximations such as Sinkhorn iterations [1]. These approaches have produced strong results, but they also introduce either unstable adversarial dynamics, computationally expensive iterative sampling, or nontrivial matching machinery.

This report explores a deliberately small alternative. SCGN trains a generator $G_\theta(z, c)$ that maps a random vector z and a continuous condition embedding c directly to an image. Instead of asking the generator to match each training example with a fixed reconstruction target, SCGN treats each minibatch as two unordered sets: generated samples and real samples. The sets are matched through a bidirectional Chamfer objective, but the matching cost is computed in a joint space that combines image similarity and condition similarity.

The main idea is that naive pixel-space set matching is underconstrained: averaging many possible targets can produce low-contrast “gray” outputs or mode collapse. SCGN adds a hyperspherical image constraint. Both generated and real images are compared only through their normalized directions, and generated images are explicitly projected to the sphere of radius $R = \sqrt{CHW}$. This removes the origin and low-norm averages as easy solutions and forces the generator to maintain nontrivial contrast.

Our contribution is not a finished large-scale generative model. It is a compact training principle:

1. project images to a fixed-radius hypersphere;
2. define a joint image-condition cosine cost;
3. optimize a bidirectional Chamfer objective over minibatch sets;
4. generate in one forward pass without adversarial, diffusion, paired reconstruction, or explicit transport-solver machinery.

2 Method

2.1 Generator and Conditioning

Let $x \in \mathbb{R}^{C \times H \times W}$ be an image and $c \in \mathbb{R}^{d_c}$ be a continuous condition vector. SCGN uses a neural generator

$$\hat{x}_i = G_\theta(z_i, c_i), \quad z_i \sim \mathcal{N}(0, I), \quad (1)$$

where z_i is normalized before being concatenated with the condition vector. The implementation used in this report is a small convolutional upsampling network with residual blocks.

For conditional experiments we use two modes. In the first, each class label is mapped to a fixed random normalized embedding. In the second, a minimal bag-of-words text encoder is trained jointly with the generator. Each training condition contains a class token and a randomly sampled adjective token, so the condition vector is continuous and each prompt instance can be unique.

2.2 Hyperspherical Image Projection

For an image tensor x , define its flattened vector as $v(x) \in \mathbb{R}^D$, where $D = CHW$. We compare image directions by

$$\bar{x} = \frac{v(x)}{\|v(x)\|_2}. \quad (2)$$

Generated images are also projected to a fixed radius:

$$G_\theta(z, c) \leftarrow R \frac{v(G_\theta(z, c))}{\|v(G_\theta(z, c))\|_2}, \quad R = \sqrt{D}. \quad (3)$$

This projection is intended to prevent collapse toward the origin or toward low-variance pixel averages. It turns the image component of training into an angular matching problem.

2.3 Joint Spherical Chamfer Objective

Given a minibatch of generated samples $\{(\hat{x}_i, \hat{c}_i)\}_{i=1}^B$ and real samples $\{(x_j, c_j)\}_{j=1}^B$, define image and condition cosine similarities:

$$s_{ij}^{img} = \bar{\hat{x}}_i^\top \bar{x}_j, \quad (4)$$

$$s_{ij}^{cond} = \bar{\hat{c}}_i^\top \bar{c}_j. \quad (5)$$

The joint matching cost is

$$C_{ij} = (1 - s_{ij}^{img}) + \gamma(1 - s_{ij}^{cond}), \quad (6)$$

where γ controls the strength of conditional alignment.

SCGN minimizes the bidirectional Chamfer loss

$$\mathcal{L}_{SCGN} = \frac{1}{B} \sum_{i=1}^B \min_j C_{ij} + \frac{1}{B} \sum_{j=1}^B \min_i C_{ij}. \quad (7)$$

The first term asks every generated sample to find a compatible real neighbor. The second term asks every real sample to be covered by at least one generated sample. Unlike a paired reconstruction loss, the nearest-neighbor assignment is computed dynamically inside each minibatch. Unlike Sinkhorn or Hungarian matching, the objective uses only row-wise and column-wise minima.

3 Implementation

The reference implementation is intentionally short. The essential training step is:

```
sim_img = torch.mm(gen_img_norm, real_img_norm.t())
sim_cond = torch.mm(gen_cond, real_cond.t())
C = (1.0 - sim_img) + gamma * (1.0 - sim_cond)
loss = C.min(dim=1)[0].mean() + C.min(dim=0)[0].mean()
```

All experiments in this preliminary report use 32×32 images, Adam with learning rate $2 \cdot 10^{-4}$, batch size 128, latent dimension 128, condition dimension 64, and $\gamma = 5$ unless otherwise noted. The code is available at <https://github.com/intexcor/SCGN>.

4 Preliminary Experiments

4.1 Datasets

We test the prototype on MNIST, Fashion-MNIST, and CIFAR-10. MNIST and Fashion-MNIST are resized to 32×32 and treated as one-channel images. CIFAR-10 is trained as a three-channel image dataset. The results reported here are qualitative samples from the current prototype, not a full benchmark.

Table 1: Preliminary experimental settings.

Dataset	Resolution	Channels	Conditioning mode
MNIST	32×32	1	fixed continuous label embedding
MNIST	32×32	1	jointly trained synthetic text encoder
Fashion-MNIST	32×32	1	fixed continuous label embedding
CIFAR-10	32×32	3	fixed continuous label embedding

4.2 Qualitative Samples

Figure 1 shows samples produced by the current implementation. The MNIST runs show recognizable digit structure under both fixed continuous label embeddings and the synthetic text-encoder mode. Fashion-MNIST shows coherent clothing-like silhouettes. CIFAR-10 remains less mature but indicates that the same objective can be applied to natural-image RGB data.

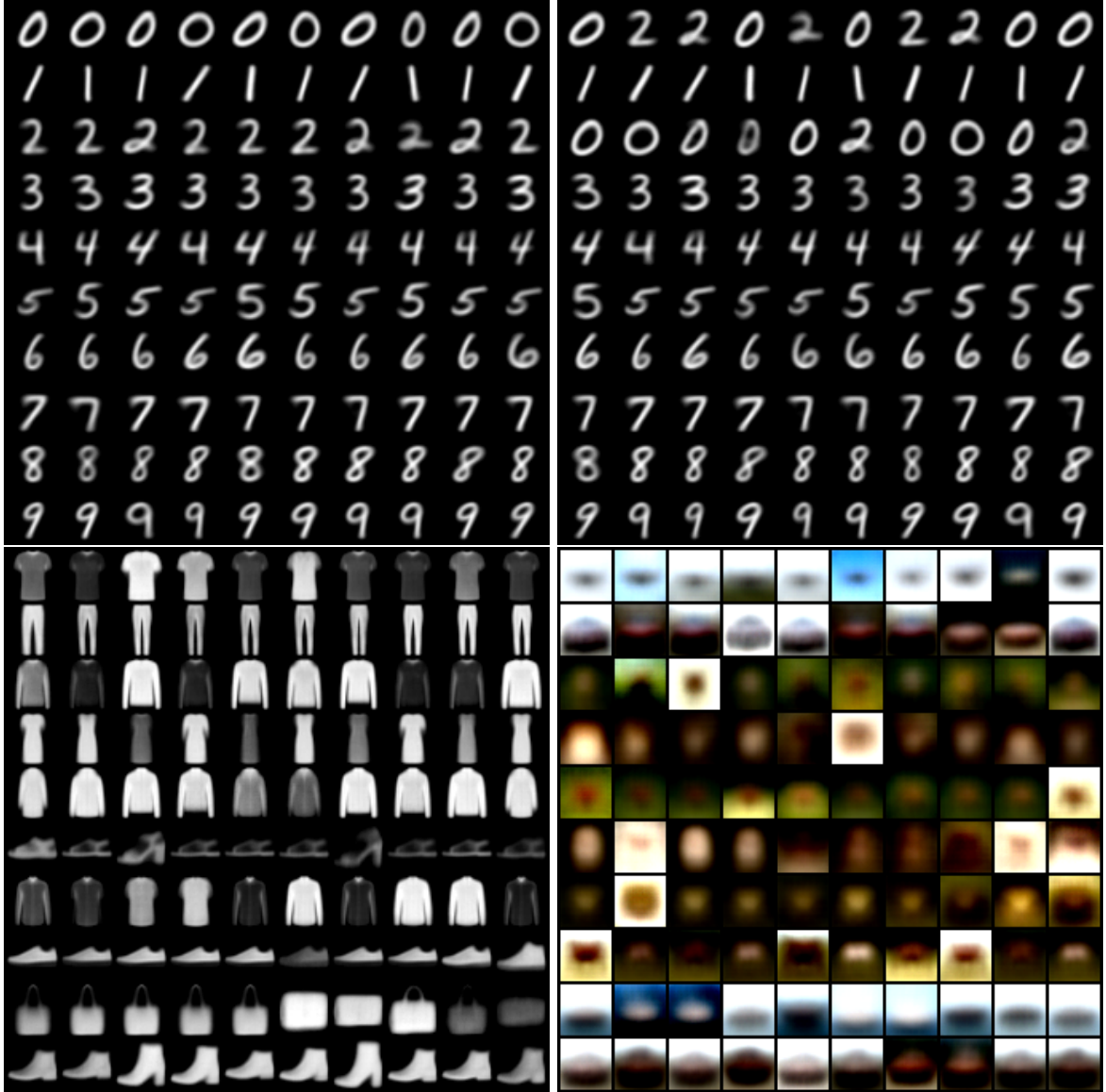


Figure 1: Preliminary SCGN samples. Top left: MNIST with fixed continuous label embeddings. Top right: MNIST with jointly trained synthetic text-encoder conditions. Bottom left: Fashion-MNIST. Bottom right: CIFAR-10.

5 Discussion

Why the sphere matters. The hyperspherical projection changes the failure modes of direct pixel-space matching. A low-contrast average image has small angular distinctiveness and is not an attractive way to cover many real samples on the sphere. The generator is therefore biased toward high-variance outputs that lie on the same norm shell as normalized real images.

Why joint matching matters. Pure image-space Chamfer matching does not identify which condition should map to which visual structure. Adding condition cosine distance to the matching cost makes generated samples compete for real samples with compatible condition vectors. In practice this acts as a routing signal without requiring an auxiliary classifier loss.

Relation to optimal transport. The objective resembles set matching, but it does not solve a balanced transport problem. There is no transport plan, entropy regularization, Sinkhorn

iteration, or Hungarian assignment. The method uses nearest-neighbor coverage in both directions as a lightweight surrogate.

6 Limitations and Future Work

This report is intentionally preliminary. The current evidence is qualitative and does not yet include FID, KID, precision-recall, classifier-based conditional alignment, or nearest-neighbor memorization tests. The text-encoder experiment uses synthetic prompts rather than a pretrained language-image model, so it should be read as evidence for continuous conditioning and routing rather than as proof of open-vocabulary text-to-image generation.

Important next steps include:

1. ablations without hyperspherical projection, without the condition term, and with one-way instead of bidirectional Chamfer;
2. sweeps over γ , batch size, latent dimension, and condition dimensionality;
3. quantitative evaluation on held-out test data;
4. nearest-neighbor analysis to measure memorization;
5. replacement of the synthetic text encoder with pretrained CLIP or T5 embeddings.

7 Conclusion

SCGN is a compact geometric prototype for one-step conditional generation. By combining hyperspherical image projection with a joint image-condition Chamfer objective, the method trains a generator without adversarial losses, diffusion sampling, paired reconstruction, or explicit transport solvers. The preliminary results suggest that this direction is worth deeper study, especially as a minimal set-matching objective for fast conditional generation.

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